Class Descriptions

Student Planner

Desktop Productivity Application

Software Engineering

1. Database module - data access layer

This module is responsible for storing and exposing application data. It contains a repository implementation that uses an SQLite database and an interface that defines the operations available to the rest of the system.

Key classes:

- IAppRepository - abstract repository interface. Defines the set of methods used by the application to manage subjects, notes, schedules, exams, language settings, and progress tracking.
- SqliteAppRepository - main data-layer implementation. Creates the database schema, runs SQL queries, and manages information stored in tables such as: subjects, class schedule, notes, events, settings.

This module is the central point of communication with the database. Other parts of the system do not run SQL directly; they use repository methods.

2. Models module - application data models

This module contains classes that represent business objects used by the user interface.

Key classes:

- SubjectData - model of a single subject. Stores information such as: subject id, name, teacher, completion status, number of absences, grading conditions, activity points, notes.

The class provides the method:

- create_from_db_dict() - converts a database record into an object used by the Kivy framework.

This model is an intermediate layer between the database and the graphical components.

3. Views module - application screens

This module implements the logic of individual application screens. Each screen inherits from Kivy's Screen class and provides a specific part of the system functionality.

Key classes:

- MySubjectsScreen - displays the list of all subjects stored in the system.
- SubjectDetailsScreen - shows detailed information about a selected subject and allows editing it.
- AddSubjectScreen - allows adding new subjects to the database.
- StartCalendar - calendar and class schedule screen; also supports adding notes to classes.
- ExamsAndColloquiumsScreen - allows adding and removing exams and colloquiums.
- ProgressTrackerScreen - lets the user monitor progress earned for subjects.
- RulesRepositoryScreen - stores grading conditions and rules that apply to subjects.

4. Widgets module - reusable UI components

This module contains independent GUI components that can be reused across multiple application screens.

Key classes:

- CountdownWidget - countdown timer for a selected event (for example, an exam session).
- DatePicker - date selection control.
- FormWidget - reusable data form.
- ProgressBar - progress bar.
- NotePopup - dialog window for notes.
- NotificationPopup - notification system.
- CalendarBrickData and LessonNote - data structures used by the calendar.

These classes define the general component behavior independently of a specific graphical implementation.

5. KivyWidgets module - graphical component implementations

This module contains widget backends prepared specifically for the Kivy framework.

Key classes:

- DashboardBrick, BrickWidget, BlueBrick - tiles that present subject information.
- CalendarWidget - graphical calendar handling.
- KivyDatePickerBackend - date picker implementation.
- KivyProgressBarBackend - progress bar implementation.
- KivyNotePopupBackend - note popup implementation.
- SessionPanel - panel that presents exam session information.
- CustomButtonWidget and CustomToggleButtonWidget - custom button components.

This module is the presentation layer between abstract widgets and the Kivy framework.

6. Style module - styles and themes

This module is responsible for the look of the application.

Key classes:

- ThemeManager - central management of the application theme.
- ButtonStyle, BasicButtonStyle - button style configuration.
- ButtonState - button state handling.
- Shapes - shape definitions for UI elements.

7. Main application module

The main application class is:

- StudentPlannerApp

This class is responsible for:

- application initialization
- creating the data repository
- loading screens
- managing translations (Polish / English)
- handling navigation between screens
- initializing the session countdown and the visual theme

Additionally, screenHandler.py acts as the screen registry, mapping screen names to the corresponding classes and .kv files.